

Responsibility-aware Strategic Reasoning in Probabilistic Multi-Agent Systems

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Outline

- 1 Motivations and overview
- 2 Parametric Model of Stochastic Game
- 3 Specification of Quantified Responsibility with PATL+R
- 4 Finding Stable Joint Plans
- 5 Conclusion

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Motivations and overview

Motivations

- In MASs, decision-making processes focus on optimising actions based solely on agents' goals and relevant payoffs.
- In human societies, **responsibility** extends beyond these considerations, requiring the ability to reason about and evaluate the consequences of actions.
- It is required to introduce mechanisms to enable **responsibility-aware strategic decision-making**:
 - ▶ Formalising and reasoning **quantified responsibility** in the MAS context.
 - ▶ Strategic reasoning with **trade-off** of **responsibility** and **payoff**.
 - ▶ Enable trustworthy AI systems.

Motivations and overview

What do we do?

- A parametric model PSMAS that captures both the uncertainty and strategic interactions in multi-agent systems of such a setting.
- Formal notions of causal responsibility and their degrees
- The PATL+R logic as a language for specifying properties that encompass quantitative responsibilities and temporal objectives.
- Parametric model checking PATL+R formulae against PSMAS.
- Apply the Nash equilibrium (NE) as the solution concept to compute stable plans.

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Parametric Model of Stochastic Game

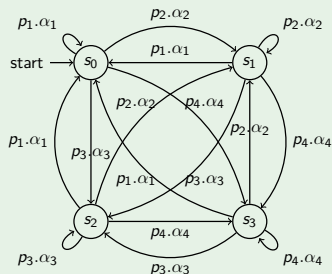
Parametric stochastic multi-agent system (PSMAS)

PSMAS: $\mathcal{M} = (\text{Ag}, S, s^0, V, \Delta, \text{Ap}, L)$, where:

- $\text{Ag} = \{1, \dots, n\}$ is a finite set of *agents*;
- S is a finite non-empty set of *states*;
- $s^0 \in S$ is the *initial state*;
- $\text{Act} = \{a_1, a_2, \dots, a_m\}$ is a non-empty finite set of actions;
- $V^i = \{x_{i,1}, \dots, x_{i,m}\}$ is a finite set of variables (**parameters**) over $\mathbb{R}^m$ for each $i \in \text{Ag}$;
- $\Delta : S \times \text{Act}^{\text{Ag}} \times S \rightarrow \mathcal{F}_V$ is the *probabilistic transition function*;
- Ap is a finite set of *atomic propositions*;
- $L : S \rightarrow 2^{\text{Ap}}$ is the *state labelling function*.

Parametric Model of Stochastic Game

Example: catching balls with parametric probabilistic transitions



$$Ag = \{A_1, A_2\}$$

$$Act = \{Catch, Skip\}$$

$$Ap = \{caught_1, caught_2\}$$

$$S = \{s_0, s_1, s_2, s_3\}$$

$$V^1 = \{x_{1,1}, x_{1,2}\} \triangleq \{x_1, 1 - x_1\}$$

$$V^2 = \{x_{2,1}, x_{2,2}\} \triangleq \{x_2, 1 - x_2\}$$

$$s_0 = dropped \triangleq \neg caught_1 \wedge \neg caught_2$$

$$s_1 = collision \triangleq caught_1 \wedge caught_2$$

$$s_2 = score_1 \triangleq caught_1 \wedge \neg caught_2$$

$$s_3 = score_2 \triangleq \neg caught_1 \wedge caught_2$$

Degree of Causal Active Responsibility

Causal Active Responsibility (CAR)

Player i bears CAR if her actions **actively** cause the outcome: if keeping i 's actions fixed, the other agents could not avoid the outcome by choosing different actions.

Definition: Degree of CAR

Given $\mathcal{M}$, the degree of the responsibility of $i \in \text{Ag}$ bearing CAR for outcome φ under joint plan π at state s is defined as:

- the ratio of the probability of behaviours following $\text{Plan}_{\pi}^{\langle\{i\}\rangle}(s)$ and leading to ω , relative to that of all possible behaviours leading to φ
— *if the occurrence of φ is avoidable*;
- and 0 — *otherwise*.

Degree of Causal Passive Responsibility

Causal Passive Responsibility (CPR)

Player i bears CPR if her actions **passively** cause the outcome: if keeping all other agents' actions fixed, i could avoid the outcome by choosing different actions.

Definition: Degree of CPR

Given $\mathcal{M}$, the degree of the responsibility of i bearing CPR for outcome φ under joint plan π at state s is defined as:

- the ratio of the probability of the behaviours following $\text{Plan}_{\pi}^{\langle \text{Ag} \setminus \{i\} \rangle}(s)$ and not leading to φ relative to that of all possible behaviours violating φ — *if the occurrence of φ following π is achievable*;
- and 0 — *otherwise*.

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Specification of Quantified Responsibility: PATL+R

Syntax: PATL+R

Let $\mathcal{M} = (\text{Ag}, S, s^0, V, \Delta, \text{Ap}, L)$.

$$\phi ::= a \mid \neg\phi \mid \phi \wedge \phi \mid \langle A \rangle \mathbf{P}_{\bowtie p}[\psi] \mid \langle A \rangle \mathbf{R}_{\bowtie q}[\Diamond_{\leq k} \phi] \\ \mid \langle A \rangle \mathbf{D}_{\bowtie d}[\text{CAR}_{i,\pi}(\psi)] \mid \langle A \rangle \mathbf{D}_{\bowtie d}[\text{CPR}_{i,\pi}(\psi)]$$

$$\psi ::= \bigcirc\phi \mid \phi \mathbf{U}_{\leq k} \phi$$

Theorem

Model checking PATL+R formula is in PSPACE.

We then present an approach to **synthesise** joint strategies **satisfying an outcome** specified in PATL+R while **optimising** the share of expected CR and reward...

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Finding Stable Joint Plans

The Idea

Game-theoretic parametric model checking to find optimised plan (Nash Equilibrium) to balance responsibility and payoffs.

Computing an NE joint plan π satisfying φ

1. Compute the set of strategy profiles satisfying the formula φ :
 - ▶ apply parametric model checking techniques on $\mathcal{M}$ to compute rational valuation functions over parameters V^i for each agent i ,
 - ▶ representing strategic transition probabilities in mixed strategies over each action $a \in \text{Act}$ in each state s .
2. Restrict this set of strategies to a subset of strategies that are NE, by:
 - ▶ formulating a set of equations corresponding to NE, and
 - ▶ solving them w.r.t. the set of strategies from step 1.

Finding Stable Joint Plans

- A plan π^i is a **best response** if it is optimal i.e., i cannot improve its payoff.
- A joint plan π is considered a mixed **Nash equilibrium** (NE) if it belongs to the best response sets for all i .

Definition: Nash Equilibrium (NE)

Given $\mathcal{M}$, for each i , a plan π^i is a **best response** w.r.t. **utility functions** if it is the set :

$$u\mathbf{BR}_i(\pi^{\text{Ag} \setminus i}) \triangleq \{\pi^i \mid \max_{\pi^i} (\lambda(\mathcal{V}_{\pi^i, \pi^{\text{Ag} \setminus i}}^i(s^0), \mathcal{R}_{\pi^i, \pi^{\text{Ag} \setminus i}}^i(s^0)))\}$$

A joint plan π is considered a mixed **Nash equilibrium** (NE) if it belongs to the best response sets $\pi \in u\mathbf{BR}_i(\pi^{\text{Ag} \setminus i})$ for all $i \in \text{Ag}$ in $\mathcal{M}$.

Theorem

Computing NE joint plan satisfying PATL+R formula φ can be done in PSPACE.

Finding Stable Joint Plans

Example: BR payoff and CR

Consider $\varphi = \langle A_1, A_2 \rangle \diamond_{\leq 2} (\text{collision} \vee \text{dropped})$, $\pi = \{(\text{Catch}_1 \text{Skip}_2)(\text{Skip}_1 \text{Skip}_2)\}$:

$$\begin{aligned}\mathbb{E}(\wp^{A_1}(\llbracket \varphi \rrbracket)) &= 2(1 - x_1)(1 - x_2) + 2(1 - x_1)x_2^2x_1 \\ &\quad + x_1^2(1 - x_2)x_2 + 2(1 - x_1)x_2(1 - x_2) \\ &\quad + 2x_1(1 - x_2)^2(1 - x_1) + x_1x_2\end{aligned}$$

$$\mathbb{P}(\mathcal{L}_\varphi) = 1 + 4x_1x_2^2 - 4x_1^2x_2^2 - x_1^2 - x_2^2 - 2x_1x_2 + 4x_1^2x_2$$

$$\mathbb{D}_{\mathcal{M}}^A(s_0, [\text{CAR}_{A_1, \pi}(\varphi)]) = \frac{(1 - x_1)(x_2^2x_1) + (1 - x_1)(1 - x_2)}{\mathbb{P}(\mathcal{L}_\varphi)}$$

$$\mathbb{D}_{\mathcal{M}}^A(s_0, [\text{CAR}_{A_2, \pi}(\varphi)]) = \frac{x_1x_2(x_2 - x_1x_2 + 1)}{\mathbb{P}(\mathcal{L}_\varphi)}$$

Finding Stable Joint Plans

Example: NE

by solving equations implied by the NE conditions:

$$\begin{cases} 2(1 - x_2) + 4x_2(1 - x_2) = x_2 + 2(1 - x_2)x_2 \\ 2(1 - x_1) = x_1 + 8(1 - x_1)x_2 \\ 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 2 \end{cases}$$

we get:

$$x_1 = \sqrt{17}/4 - 1/4 \approx 0.78, \quad x_2 = 2/3$$

i.e., when A_1 takes action of `skip` with probability of 0.78 and A_2 takes action of `skip` with probability of $2/3$, a responsibility-aware plan is obtained.

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Conclusion and Future Works

Summary

- An approach to multi-agent strategic reasoning with responsibility-aware agents, by introducing
 - ▶ logic PATL+R, to reason about (quantified) causal responsibility and temporal properties,
 - ▶ model-checking PATL+R formulae against a parametric model of probabilistic MAS.
- An approach to synthesise NE joint strategies/plans that satisfy some PATL+R property to find optimised plans.

Future work

- Finite-memory strategy settings
- Extension to more expressive logic, s.a., PSL
- What can be done if no NE joint plan satisfies the required property?